

TEACHING STATEMENT

by LILI JU

I chose Mathematics as the major during my undergraduate study (1991-1995) in Wuhan University in China partially because I was deeply attracted by mathematics when I was very young. My high-school mathematics teacher was an excellent teacher who first led me to learn the beauty of mathematics and made me believe I also had the talent to discover it. From him I also learned to appreciate teaching and value the hard work it requires. Since then I have met many many other great mathematics professors later in my life, all in their own way inspire me with their love for the profession and teach me the patience and joy of sharing knowledge.

Regular Teaching Experiences

I went to the Iowa State University to pursue a Ph.D. degree in August 1998 after obtaining a M.S. Degree in Computational Mathematics from the Institute of Computational Mathematics of Chinese Academy of Sciences. During my first-year graduate study at Iowa State, I was a fresh Teaching Assistant working in the mathematics help room, and responsible for answering problems from students on all introductory-level mathematical courses. From January 2001 to December 2001, I worked as a Teaching Assistant for two elementary undergraduate courses – *College Algebra* and *Calculus I*. These courses are part of the core requirements for most mathematics, sciences and engineering majors, and consequently have very high enrollments. My duties ranged from conducting recitations, to arranging and grading homework, quizzes and exams, to holding office hours for answering students' questions.

In August 2004, I joined the University of South Carolina (USC) at Columbia as an Assistant Professor of Mathematics after finishing a two-year's Industrial Postdoctoral training in the Institute for Mathematics and Its Applications (IMA) at the University of Minnesota. Since then I have taught various regular courses for both undergraduate and graduate students. At the undergraduate level, I taught 7 different courses: *Calculus I* (MATH 141), *Calculus II* (MATH 142), *Vector Calculus* (MATH 241), *Elementary Differential Equations* (MATH 242), *Ordinary Differential Equations* (MATH 520), *Numerical Linear Algebra* (MATH 526), and *Numerical Analysis* (MATH 527); at the graduate level, I taught 5 different courses: *Numerical Linear Algebra* (MATH 706), *Foundations of Computational Mathematics I* (MATH 708), *Numerical Analysis I&II* (MATH 726&727), and *Scientific Computing I* (MATH 728J). I also had undergraduate and graduate student research credit courses such as Directed Readings & Research (MATH 798), Thesis Research (MATH 799), Dissertation Research (MATH 899), and Senior Thesis/Project (SCHC 499A). For details of my teaching experiences and corresponding student evaluations at South Carolina, see Table 1. I enjoy teaching elementary classes in all areas of mathematics for undergraduate students although I am primarily interested in courses in computational and applied mathematics (my research field) at the graduate level.

Research Supervision Experiences

In addition to regular teaching, another important task of my duties at South Carolina is research supervision to graduate students, undergraduates, and postdocs. I was ever the member of both graduate and undergraduate advisory committees providing help to students in taking courses and to the department in recruiting new students and developing new courses.

I so far have finished supervising two Ph.D. students

- Xiao Xiao (Ph.D., Mathematics, 2012/5, Dissertation title: "*Covolume-upwind Finite Volume Approximations for Linear Elliptic Partial Differential Equations*")
- Li Tian (Ph.D., Mathematics, 2009/8, Dissertation title: "*Error Estimates for Finite Element/Volume Approximations of Dissipative Partial Differential Equations on Surfaces*")

TERM	NUMBER	COURSE TITLE	CRS.	ENROLL.	STU. EVAL.
Spring 2013	MATH 242	Elementary Ordinary Equations	3	30	
	MATH 709	Foundations of Comput. Math. II	3	3	
	MATH 799	Thesis Research	3	1	-
Fall 2012	MATH 241	Vector Calculus	3	45	
	MATH 708	Foundations of Comput. Math. I	3	4	
	MATH 798	Directed Readings and Research	1	1	-
Summer 2012	MATH 798	Directed Readings and Research	1	1	-
Spring 2012	MATH 142	Calculus II	4	63	4.40/5
	MATH 526	Numerical Linear Algebra	4	43	3.94/5
	MATH 899	Dissertation Research	1	1	-
Fall 2011	MATH 141	Calculus I	4	63	4.05/5
	MATH 708	Foundations of Comput. Math. I	3	8	4.43/5
	MATH 899	Dissertation Research	3	1	-
Summer 2011	MATH 899	Dissertation Research	1	1	-
Spring 2011	SCHC 499A	Senior Thesis/Project	3	2	-
	MATH 899	Dissertation Research	3	1	-
	MATH 899	Dissertation Research	3	1	-
Fall 2010	MATH 899	Dissertation Research	3	1	-
Summer 2010	MATH 799	Thesis Research	1	1	-
	MATH 899	Dissertation Research	1	1	-
	MATH 141	Calculus I	4	55	4.61/5
Spring 2010	SCHC 499A	Senior Thesis/Project	3	1	-
	MATH 527	Numerical Analysis	3	9	4.60/5
	MATH 799	Thesis Research	3	1	-
Fall 2009	MATH 899	Dissertation Research	3	1	-
	MATH 526	Numerical Linear Algebra	4	20	4.31/5
	MATH 242	Elementary Differential Equations	3	36	4.74/5
Spring 2009	MATH 799	Thesis Research	3	1	-
	MATH 899	Dissertation Research	3	1	-
	MATH 526	Numerical Linear Algebra	4	17	4.27/5
Fall 2008	MATH 899	Dissertation Research	3	1	-
	MATH 899	Dissertation Research	1	1	-
	MATH 899	Dissertation Research	1	1	-
Summer 2008	MATH 899	Dissertation Research	1	1	-
Spring 2008	MATH 706	Numerical Linear Algebra	3	6	4.78/5
	MATH 899	Dissertation Research	3	1	-
	MATH 899	Dissertation Research	3	1	-
Fall 2007	MATH 520	Ordinary Differential Equations	3	24	4.54/5
	MATH 899	Dissertation Research	3	1	-
	MATH 899	Dissertation Research	3	1	-
Summer 2007	MATH 798	Directed Readings & Research	1	1	-
	MATH 799	Thesis Research	1	1	-
	MATH 727	Numerical Analysis II	3	4	4.00/4
Spring 2007	MATH 799	Thesis Research	3	1	-
	MATH 799	Thesis Research	3	1	-
	MATH 526	Numerical Linear Algebra	4	17	3.64/4
Fall 2006	MATH 726	Numerical Analysis I	3	7	4.00/4
	MATH 141	Calculus I	4	57	2.11/4
	MATH 526	Numerical Linear Algebra	4	19	3.38/4
Spring 2006	MATH 526	Numerical Linear Algebra	4	19	3.38/4
Fall 2005	MATH 728J	Scientific Computing I	3	5	4.00/4
Spring 2005	MATH 526	Numerical Linear Algebra	4	23	3.64/4
Fall 2004	MATH 241	Vector Calculus	3	46	3.21/4

Table 1: Courses taught at the University of South Carolina and student evaluations.

and three M.S. students

- Jing Liu (M.S., Mathematics, 2010/8, Thesis title: *“Construction of Centroidal Voronoi Tessellations Using a Conjugate Gradient Method Based on Trust Regions”*)
- Xiao Xiao (M.S., Mathematics, 2010/8, Thesis title: *“Over-relaxation Lloyd Method for Com-*

puting Centroidal Voronoi Tessellations”)

- Wensong Wu (M.S., Mathematics, 2007/8, Thesis title: “*Adaptive Finite Volume Methods for Convection-Diffusion Equations with Mesh Optimization*”)

In particular, Dr. Tian was one of the four winners of *USC Dean’s Award for Excellence in Graduate Study* in 2009. I am currently advising two graduate students:

- Yu Cao (Ph.D., Computer Science and Engineering, co-supervised with Prof. Song Wang from CSE department at USC). Yu mainly works on *EWCVT-based image segmentation methods for 3D super-alloy grain segmentation* and is expected to graduate in the summer of 2013.
- Rosalia Tatano (M.S., Mathematics). Rosalia works on *covolume-upwind finite volume methods for second-order time-dependent problems* and is expected to graduate in the summer of 2013.

During the period of 2010/8 to 2011/2 I supervised a visiting Ph.D. student, Yang Li, from Shandong University in China who was supported by the China Scholarship Council. Yang Li worked on *development and analysis of numerical methods for solving backward and forward-backward stochastic differential equations*. Another visiting Ph.D. student Yongqiang Ren from China is currently advised by Prof. Hong Wang and me at USC. In addition, I have been member of the Doctoral Dissertation Committees for 11 Ph.D. students since 2004.

I also have been actively involved in postdoctoral training. From 2008/9 to 2010/3, Dr. Huai Zhang was supported by my research grants and worked with me on *parallel finite element 3D nonlinear Stokes ice-sheet modeling*.

In 2006, I was the faculty advisor of USC teams for the Mathematical Contest in Modeling (MCM), consisting of undergraduate students. In that year, the USC team with William Kay and Raleigh Youmans as members won the MCM *Honorable Mention* award under my supervision.

Teaching Philosophy

I do believe that a good teacher is not only to teach effectively what students need to study, but also to motivate them to learn more. Mathematics can be really intimidating for some students, while others may find slow-paced classes boring, so there should be enough flexibility in the difficulty level allowing for a healthy balance. Additionally, mutual respect is also very important and is a good way to maintain the teaching quality and students’ interests to the courses. I believe students tend to learn better and are more open to new ideas when they are surrounded by a friendly, learner-centered environment, where they feel apt and their opinions are well valued. It is important that they feel comfortable to ask questions and are encouraged by the instructor to do so.

To summarize, I believe that my teaching and research experience and potential have prepared me to take on the challenges of being an outstanding researcher as well as an excellent teacher. I really enjoyed communicating with students and watering their thirst for knowledge. I am willing to devote myself to mathematics teaching since I find teaching to be a very rewarding experience.